

Technology Stack

Date	29 June 2026
Team ID	xxxxxx
Project Name	Rising Waters
Maximum Marks	2 Marks

Technology Stack Details :

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	<i>HTML, CSS, Bootstrap, JavaScript</i>	<i>Used to develop a responsive and user-friendly interface for entering weather parameters and displaying flood prediction results.</i>

2	Backend / Server-Side	<i>Python, Flask</i>	<i>Used to process user inputs, integrate machine learning models, and generate flood predictions through the web application.</i>
3	Database / Data Storage	<i>CSV Files, Pandas DataFrames</i>	<i>Used to store and process historical weather and flood datasets required for model training and prediction.</i>
4	Cloud / Hosting / Deployment	<i>IBM Cloud</i>	<i>Used for scalable deployment and remote accessibility of the flood prediction application.</i>
5	Version Control & CI/CD	<i>Git, GitHub</i>	<i>Used for source code management, version tracking, and maintaining project versions efficiently.</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools	<i>Scikit-learn, XGBoost, NumPy, Pandas, Jupyter Notebook, PyCharm.</i>	<i>Used for data preprocessing, machine learning model development, model evaluation, and experimentation.</i>